

Nuclear Norm-Regularized Bayesian Matrix Completion

Calvin Tolbert
School of Operations Research and Information Engineering
Cornell University
cmt238@cornell.edu

September 21, 2026

Abstract

Matrix completion, the problem of estimating missing entries in a matrix from noisily observed ones, underlies a diverse array of problems such as recommender systems and counterfactual outcome estimation in panel data. Many algorithms address the problem using regularized least squares, often with the nuclear norm as a regularizer, but this method yields a point estimate with no built-in uncertainty quantification. A Bayesian formulation is a natural alternative, and if the noise variance is known, the nuclear norm-based prior yields a log-concave posterior. Unfortunately, in practice, the noise variance will not be known *a priori*, so for a fully Bayesian approach, a prior must be imposed on it. We give the first sampler for this model with an explicit non-asymptotic guarantee: polynomial in the matrix dimensions and in the reciprocal of the target accuracy. Our technique is to discretize the distribution of the noise precision onto a grid and build a categorical posterior via thermodynamic integration. This extension is not specific to matrix completion and may be useful in other non-log-concave sampling problems where the non-log-concavity is restricted to a single variable and the joint distribution of the remaining variables is nonsmooth. Our contribution is a feasibility result: we show that a polynomial-time Bayesian sampler for this model exists at all, and the resulting complexity, while polynomial, is not intended as a deployable algorithm at current problem scales.

1 Introduction

Matrix completion and low-rank matrix estimation have gained significant traction in modern data science due to their broad applicability in domains such as recommender systems, collaborative filtering, causal inference, and imaging. These problems revolve around recovering an underlying low-rank structure from noisy and incomplete observations. Traditional approaches are predominantly optimization-based, formulating the problem as regularized least squares with a nuclear-norm penalty to induce low-rankness (see, for example, [Negahban and Wainwright \(2010\)](#), [Athey et al. \(2021\)](#), [Chen et al. \(2020b\)](#)). Such methods typically focus on point estimates and provide limited insight into uncertainty quantification. The Bayesian framework offers a probabilistic interpretation of regularization and full posterior distributions, facilitating credible intervals and more principled uncertainty analysis, but Bayesian methods remain underutilized here due to computational complexity and the challenges of posterior sampling in high-dimensional latent spaces.

This paper addresses one key obstacle to putting a fully Bayesian nuclear-norm-type prior on a rigorous, non-asymptotic footing: making such a prior log-concave requires knowing the noise precision, which is rarely available in practice; existing treatments either fix it in advance ([Chen et al. \(2020b\)](#)), use cross-validation ([Athey et al. \(2021\)](#)), or integrate it out via Gibbs sampling with no non-asymptotic guarantee on the result ([Cui and Gorodetsky \(2024\)](#)). We instead place

a prior on the precision and give an algorithm, together with an explicit non-asymptotic total-variation bound, for sampling from the joint posterior over the precision and the low-rank factor. A naïve version of this marginalization — estimating the marginal likelihood at each candidate precision by importance-reweighting samples drawn from the untilted prior — requires a number of samples growing exponentially in the number of observations, since the importance weights degrade as the likelihood tilt moves away from the prior. We replace this with a thermodynamic-integration estimator that samples only from the correctly tilted measure at each precision value, removing the exponential dependence.

The natural nuclear-norm-plus-Frobenius-norm prior is log-concave but not smooth, and non-asymptotic sampling theory has mostly been developed for smooth targets. Composite potentials of the form (smooth) + (nonsmooth, Lipschitz) are not entirely uncharted: [Mou et al. \(2022\)](#) give an efficient Metropolis–Hastings scheme for exactly this class. The price of removing this oracle requirement is worse conditioning: our mixing bound depends exponentially on the ratio of the non-smooth term’s Lipschitz constant to the smooth term’s curvature, whereas [Mou et al. \(2022\)](#)’s rate matches smooth-target samplers up to the condition number. Their algorithm, however, depends on an efficient proximal sampling oracle for the nonsmooth term. The nuclear norm has a closed-form proximal operator, but our proof will apply to a general convex and Lipschitz function. We do this by extending a different tool, the isoperimetric-profile mixing-time framework of [Andrieu et al.](#), developed for smooth strongly log-concave targets sampled by random-walk Metropolis. Random-walk Metropolis needs only potential evaluations, not a proximal sampling step, so this extension applies directly to the nuclear-norm setting; the resulting mixing bounds are polynomial in dimension and logarithmic in the target accuracy. This extension is not specific to matrix completion and may be of independent interest for nonsmooth log-concave sampling problems more broadly, wherever an efficient proximal sampling oracle for the nonsmooth term is unavailable.

We use this model as a concrete testbed for a more general question: can random-walk Metropolis get explicit non-asymptotic mixing bounds on smooth-plus-nonsmooth-Lipschitz targets without a proximal oracle? How about if the joint distribution of all the variables is not strongly log-concave?

Our contributions are: (1) a thermodynamic-integration-based marginalization scheme for the noise precision with an explicit non-asymptotic total-variation guarantee, avoiding the exponential sample complexity of the naïve importance-weighted estimator; and (2) showing via a direct growth-bound computation that the isoperimetric-profile mixing-time analysis of random-walk Metropolis found in [Andrieu et al. \(2024\)](#) extends to the case of smooth-plus-nonsmooth-Lipschitz composite potentials. Both contributions serve a single strategy: at no point do we sample from a continuous distribution that is not strongly log-concave.

We introduce some notation and parameters here, but we save the model itself for later.

- $\mathbf{Y} \in (\mathbb{R} \cup \{\square\})^{n_1 \times n_2}$ is the (generally incomplete) matrix of observations. We will assume without loss of generality that $n_1 \leq n_2$. Unobserved entries are denoted by $\square$, where $0\square = 0$. Our goal is to estimate this matrix.
- $\mathbf{L} \in \mathbb{R}^{n_1 \times n_2}$ is the true latent-factor matrix.
- $\mathbf{E} \in \mathbb{R}^{n_1 \times n_2}$ is the matrix of residuals, which are assumed to be i.i.d. Gaussians with mean 0 and precision τ , noting that the precision of a random variable is the reciprocal of its variance.
- $\mathbf{\Omega} \in \{0, 1\}^{n_1 \times n_2}$ is the sampling mask where each entry is 1 if the corresponding entry of $\mathbf{Y}$ is real and 0 if it is $\square$. We will often interact $\mathbf{\Omega}$ with $\mathbf{Y}$ using the entry-wise Hadamard product, denoted $\circ$, as in $\mathbf{Y} \circ \mathbf{\Omega}$.
- N is the number of observations, or the number of entries of $\mathbf{\Omega}$ which are equal to one.

2 Summary of existing models

The field of matrix completion has a rich history, going back to the seminal papers of [Keshavan et al. \(2010\)](#); [Keshavan and Montanari \(2010\)](#); [Candès and Tao \(2010\)](#) and the more general paper of [Negahban and Wainwright \(2010\)](#), with applications ranging from causal inference in panel data as in [Bai and Ng \(2021\)](#); [Agarwal et al. \(2021\)](#); [Athey et al. \(2021\)](#) to recommender systems as in [Koren et al. \(2009\)](#). The nuclear-norm regularization approach, penalizing model complexity, allows for a broader class of models than the synthetic control approach of [Abadie et al. \(2010\)](#) while still preferring simple models over more complicated ones. Some examples of this approach are [Negahban and Wainwright \(2010\)](#); [Chen et al. \(2020b\)](#); [Athey et al. \(2021\)](#); [Cui and Gorodetsky \(2024\)](#). [Chen et al. \(2020b\)](#) uses an estimator of the form

$$\arg \min_{\mathbf{L}, \mathbf{S}} \frac{1}{2} \|(\mathbf{L} + \mathbf{S} - \mathbf{Y}) \circ \boldsymbol{\Omega}\|_{\text{F}}^2 + \frac{1}{\lambda} \|\mathbf{L}\|_* + \frac{1}{\lambda'} \|\mathbf{S}\|_{1,1}$$

where $\mathbf{S}$ is a sparse matrix of extreme outliers, developing a theory to explain its strong empirical performance. The authors assume that the entries of the residual matrix $\mathbf{E}$ are symmetric and sub-Gaussian in addition to being i.i.d. with mean zero, and that the entries are missing completely at random. In addition, they impose incoherence assumptions on $\mathbf{L}$ to ensure that the singular vectors are not too close, and they bound the smallest singular value in terms of the sub-Gaussian norm of the residuals. These assumptions, though necessary for the proof, are difficult to verify in practice. Another recent paper using nuclear-norm regularization is [Athey et al. \(2021\)](#), which serves to broaden the scope of other matrix completion papers in the panel data literature by allowing time-series dependency, so that once a unit is treated, it can stay treated for the duration of the dataset. Under our Bayesian framework, we will generalize this even further to allow for arbitrary dependencies in treatment status. The authors also propose a cross-validation approach for choosing λ . Under a Bayesian interpretation of the regularization term, many of these assumptions are no longer necessary.

3 Regularization as Bayesian inference

3.1 Regression

Bayesian interpretations of regularized regression models are almost as old as the models themselves. [Tibshirani \(1996\)](#), the paper introducing the statistics world to LASSO regularization, points out that regularized least-squares estimators of the form

$$\arg \min_{\boldsymbol{\beta}} \frac{\tau}{2} \|(\mathbf{X} \cdot \boldsymbol{\beta} - \mathbf{Y}) \circ \boldsymbol{\Omega}\|_{\text{F}}^2 + \frac{1}{\gamma} \|\boldsymbol{\beta}\|_1$$

can be viewed as the maximum *a posteriori* (MAP) estimator of regression coefficients with priors

$$\beta_k \stackrel{\text{i.i.d.}}{\sim} \text{Laplace}(0, \gamma) \quad \forall k \in [p]$$

if the precision of the residuals is known to be τ , there are no latent factors, and the Central Limit Theorem can be used to give a likelihood function. This is because the objective function above is the posterior negative log-pdf.

3.2 Latent factor models

We will focus on the case of nuclear norm-regularized regression. For a matrix $\mathbf{L}$, the nuclear norm $\|\mathbf{L}\|_*$ is the sum of the singular values of $\mathbf{L}$, and this regularization term penalizes high-rank matrices since these will have many nonzero singular values. Reparameterizing slightly, the estimator of [Chen et al. \(2020b\)](#) can be written as

$$\arg \min_{\mathbf{L}, \mathbf{S}} \frac{\tau}{2} \|(\mathbf{L} + \mathbf{S} - \mathbf{Y}) \circ \boldsymbol{\Omega}\|_{\text{F}}^2 + \frac{1}{\lambda} \|\mathbf{L}\|_* + \frac{1}{\lambda'} \|\mathbf{S}\|_{1,1},$$

where $\|\cdot\|_{1,1}$ denotes the entry-wise 1-norm. This is also a regularized regression estimator, and it can be interpreted as the MAP estimator if we have priors

$$p(\mathbf{L}) \propto \exp(-\|\mathbf{L}\|_*/\lambda), \quad p(\mathbf{S}) \propto \exp(-\|\mathbf{S}\|_{1,1}/\lambda').$$

The prior on $\mathbf{S}$ has the convenient interpretation that

$$S_{ij} \stackrel{\text{i.i.d.}}{\sim} \text{Laplace}(0, \lambda') \quad \forall i \in [m], j \in [n].$$

Taking this Bayesian interpretation seriously, rather than treating it as a computational convenience, has a further payoff. Frequentist analyses of the nuclear-norm estimator typically require incoherence of $\mathbf{L}$'s singular vectors, a minimum sampling rate on $\boldsymbol{\Omega}$, or entries missing completely at random to control the point estimator's worst-case error. These conditions are not needed here: our sampler's guarantee holds for the posterior induced by μ for any $\boldsymbol{\Omega}$, because it is a statement about how well the sampler reaches a well-defined target distribution, not about how close that target's mode is to the truth. Assumptions that exist only to bound a frequentist estimator's error are, in this framing, simply not assumptions our guarantee needs.

4 Our prior

For our proposed distribution, we will impose priors on latent factors using both the nuclear norm and the Frobenius norm:

$$\mu(\mathbf{L}) \propto \exp\left(-\frac{1}{\lambda} \|\mathbf{L}\|_* - \frac{1}{B} \|\mathbf{L}\|_{\text{F}}^2\right).$$

The nuclear-norm component makes sense for shrinkage purposes to induce near-low-rankness, as documented in the earlier sections of this paper. The Frobenius-norm term has the computational advantage of making μ *strongly* log-concave rather than simply log-concave, as well as the theoretical advantage of providing entrywise shrinkage in addition to the rank shrinkage of the nuclear norm. A user who is solely interested in low-rankness and only sees the Frobenius term as a vehicle for computational tractability can simply set B very large compared to λ .

One obvious shortcoming of the simple Bayesian interpretation of the Frobenius norm is the unrealistic assumption that the residual precision is known. Unlike virtually every paper except [Cui and Gorodetsky \(2024\)](#), we will add a prior on the precision. A natural choice is the improper Jeffreys prior, $p(\tau) \propto 1/\tau$, which is uninformative by design. The Jeffreys prior for a parameter q is defined in terms of its Fisher information matrix:

$$p(q) \propto |I(q)|^{1/2}.$$

This definition is useful because of its invariance to monotone transformation. For example, the Jeffreys prior pdf on the residual standard deviation is inversely proportional to the standard deviation, giving the same distribution as if the problem had been expressed in terms of τ . This prior

is improper, though, and its use would cause trouble in later sections, since it would also cause the posterior to be improper. As such, we will truncate the support of τ , giving us

$$\mu_\star(\tau) = \frac{1}{\tau \log \kappa} \mathbb{1}\{\tau_{\min} \leq \tau \leq \tau_{\max}\},$$

with $\kappa := \tau_{\max}/\tau_{\min}$. This is the prior we will use.

4.1 Random-walk Metropolis

The first task will be to sample from the prior μ on $\mathbf{L}$. μ is $2/B$ -strongly log-concave, but because of the nuclear-norm term, it is not smooth. Much recent literature (e.g., [Durmus et al. \(2022\)](#); [Durmus and Moulines \(2017\)](#); [Salim et al. \(2019\)](#); [Salim and Richtarik \(2020\)](#)) has addressed the problem of sampling from log-concave distributions where the potential is the sum of

- a smooth and strongly convex function ($\|\mathbf{L}\|_F^2$ is 2-smooth and 2-strongly convex), and
- a convex nonsmooth function with a tractable proximal operator ($\|\mathbf{L}\|_*$ is convex and

$$\text{prox}_{c\|\cdot\|_*}(\mathbf{L}) = \mathbf{U}[\mathbf{\Sigma} - c\mathbf{I}]^+ \mathbf{V}^\top$$

if $\mathbf{L} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ is the SVD).

Langevin Monte Carlo generally requires smoothness [Chewi \(2024\)](#) and converges in a number of iterations which is polynomial in the dimension and the reciprocal of the error tolerance. We will see in the appendix, though, that in the posterior case our error tolerance must be exponentially small to accommodate an exponentially large factor. This calls for an algorithm where the number of iterations to reach convergence is polynomial in the *logarithm* of the error tolerance. Though nice, the proximal Langevin sampler of [Salim et al. \(2019\)](#) does not meet this criterion. A few such high-accuracy samplers exist and have been studied, with Metropolis-adjusted Langevin ([Dwivedi et al. \(2019\)](#)) and Metropolized Hamiltonian Monte Carlo ([Lee et al. \(2020\)](#); [Chen et al. \(2020a\)](#)) being notable examples, but the one whose theory includes our setting is random-walk Metropolis as in [Andrieu et al. \(2024\)](#).

Algorithm 1 Random-Walk Metropolis for $\pi \propto \exp(-V)$

Require: step size $\sigma > 0$; number of iterations K ; initial distribution π^0

```

1: draw  $\mathbf{L}^0 \sim \pi^0$ 
2: for  $k = 0, 1, \dots, K - 1$  do
3:    $\mathbf{Z}^k \sim \mathcal{N}(\mathbf{L}^k, \sigma^2 \mathbf{I})$  ▷ proposal
4:    $U^k \sim \text{Unif}(0, 1)$ 
5:   if  $\log U^k \leq V(\mathbf{L}^k) - V(\mathbf{Z}^k)$  then ▷ symmetric proposal  $\Rightarrow$  no Hastings correction
6:      $\mathbf{L}^{k+1} \leftarrow \mathbf{Z}^k$ 
7:   else
8:      $\mathbf{L}^{k+1} \leftarrow \mathbf{L}^k$ 
9:   end if
10: end for
11: return  $\mathbf{L}^K$ 
```

4.2 Discretizing the precision

We can sample from $\mu_\star$ without trouble, but when it comes time to compute the posterior, it will be helpful to discretize. We define $\mu_\star^Q$ to be discretization of our prior on τ : we choose Q logarithmically spaced points $\tau_1, \dots, \tau_Q$ so that the prior is approximated by the uniform categorical distribution over this set. The q th discretization point is therefore the $(2q-1)/2Q$ quantile of the prior:

$$\tau_q := \tau_{\min} \kappa^{(2q-1)/2Q}.$$

5 Our posterior

Because residuals are assumed to be Gaussian, our likelihood is

$$p(\mathbf{Y}|\mathbf{\Omega}, \mathbf{L}, \tau) \propto \tau^{N/2} \exp\left(-\frac{\tau}{2} R(\mathbf{L})\right),$$

$$R(\mathbf{M}) := \|(\mathbf{M} - \mathbf{Y}) \circ \mathbf{\Omega}\|_{\text{F}}^2.$$

Defining the tilted measure

$$\mathrm{d}\rho_s \propto \exp\left(-\frac{s}{2} R(\mathbf{L})\right) \mathrm{d}\rho(\mathbf{L})$$

for a given measure ρ , we can see that our prior is $\mu = \mu_0$ and that if τ were known, the posterior would be μ_τ . However, τ is not known. How do we sample from the posterior on $\mathbf{L}$ when τ is also uncertain? A common approach to sampling from distributions with different groups of variables is Gibbs sampling, as in [Abrahamsen and Hobert \(2017\)](#); [Hobert and Geyer \(1998\)](#); [Jones and Hobert \(2004\)](#); [Román and Hobert \(2012\)](#); [Rosenthal \(1995\)](#), but general convergence rate guarantees are difficult to come by. However, if we can sample from the *marginal* posterior of the precision, we could then sample from the conditional posterior of $\mathbf{L}$. To do this, we define some new terms.

- $\nu_\star$ is the posterior measure on τ .
- $\mathbf{\Lambda}$ is the vector of marginal log-likelihoods over the τ_q 's; that is,

$$\Lambda_q := \log \mathbb{E}_{\mathbf{L} \sim \mu} [p(\mathbf{Y}|\mathbf{L}, \tau_q)].$$

- $\nu_\star^Q := \text{softmax}(\mathbf{\Lambda})$ is the categorical distribution induced by normalizing $\mathbf{\Lambda}$.
- *Thermodynamic integration* is a technique for computing marginal posteriors by integrating over a parameterized path from the prior to the posterior, which is a one-dimensional integral.
- $\tilde{\mathbf{\Lambda}}$ is the vector of approximate marginal log-likelihoods over the τ_q 's computed using thermodynamic integration as described above and detailed in [Algorithm 2](#).
- $\nu_\star^{Q, \text{TI}} := \text{softmax}(\tilde{\mathbf{\Lambda}})$ is the categorical distribution induced by normalizing $\tilde{\mathbf{\Lambda}}$.
- μ_τ is the posterior measure on $\mathbf{L}$ conditional on τ , from which we cannot sample.
- μ_τ^K is the output of [Algorithm 1](#) after K iterations starting at μ_τ^0 , and it is separated from μ_τ by $\varepsilon_\star$.

Algorithm 2 Thermodynamic-integration marginal-likelihood estimator

Require: step size $\sigma > 0$; iteration function $K(\cdot)$; grid $\tau_1 < \dots < \tau_Q$; inner sample size M

```
1:  $\hat{\Lambda}_1 \leftarrow 0$ 
2: for  $i \in \{1, \dots, Q-1\}$  do
3:   for  $j \in \{1, \dots, M\}$  do
4:      $\mathbf{L}_j^{(i)} \leftarrow \text{RWM}(\mu_{\tau_i}^0, \sigma, K(\tau_i))$ 
5:   end for
6:    $\hat{R}_M(\tau_i) \leftarrow \frac{1}{M} \sum_{j=1}^M R(\mathbf{L}_j^{(i)})$ 
7: end for
8: for  $q \in \{2, \dots, Q\}$  do
9:    $\bar{\Lambda}_q \leftarrow \sum_{i=1}^{q-1} \left( \frac{N}{2\tau_i} - \frac{1}{2} \hat{R}_M(\tau_i) \right) (\tau_{i+1} - \tau_i)$ 
10: end for
11: return  $\nu_\star^{Q, \text{TI}} \leftarrow \text{softmax}(\{\bar{\Lambda}_1, \dots, \bar{\Lambda}_Q\})$ 
```

Algorithm 3 Posterior sampler

Require: distribution $\nu_\star^{Q, \text{TI}}$

```
1: for  $i \in \{1, \dots, \text{num\_samples}\}$  do
2:    $\tau^{(i)} \sim \hat{\nu}_\star^{Q, \text{TI}}$ 
3:    $\mathbf{L}'_i \leftarrow \text{RWM}(\mu_{\tau^{(i)}}^0, \sigma, K(\tau^{(i)}))$ 
4: end for
5: return  $\{\mathbf{L}'_i : i \in [\text{num\_samples}]\}$ 
```

Our goal is now to sample from $\int \mu_\tau \nu_\star$. Define

$$\mu_\tau^0(\mathbf{L}') \propto \exp \left(-\frac{1}{B} \|\mathbf{L}'\|_{\text{F}}^2 - \frac{\tau}{2} \|(\mathbf{L}' - \mathbf{Y}) \circ \boldsymbol{\Omega}\|_{\text{F}}^2 \right).$$

The first thing we do is create $\nu_\star^{Q, \text{TI}}$ by sampling from Algorithm 2. To sample from the posterior μ_τ , we assign a number of iterations $K(\tau)$, sample τ from our constructed $\nu_\star^{Q, \text{TI}}$, and sample from $\mu_\tau^{K(\tau)}$ conditioned on that τ .

Because the smooth part of μ_t 's potential is $2/B$ -strongly convex and $(2/B+t)$ -smooth and the nuclear-norm term is $(\sqrt{n_1}/\lambda)$ -Lipschitz, and $\mathbb{E}_{\mu_t^0}[\mathbf{L}] = (1 - \frac{2}{Bt+2})\mathbf{Y} \circ \boldsymbol{\Omega}$, the error of the algorithm is as follows. Note that the notation is in terms of mixtures of probability distributions, with a conditional law of $\mathbf{L}$ given τ and a marginal law of τ written next to each other in an integrand.

Theorem 1. *If N is large and the values of the quantities in Table 1 are as in Table 2, we have we have*

$$\begin{aligned} & \text{TV} \left(\int \mu_\tau \nu_\star(d\tau), \int \mu_\tau^{K'} \mathbb{E}_{\mu^K}[\nu_\star^Q](d\tau) \right) \\ & \leq (\varepsilon_\star + \sqrt{R_1 \mathcal{W}_1}) / \sqrt{2}. \end{aligned}$$

Theorem 2. *Under the same setting as Theorem 1, we have*

$$\begin{aligned} & W_1 \left(\int \mu_\tau \nu_\star(d\tau), \int \mu_\tau^{K'} \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right) \\ & \leq \sqrt{B}(\varepsilon_\star + \sqrt{R_1 \mathcal{W}_1}), \end{aligned}$$

Table 1: Glossary of auxiliary quantities

Symbol	Role
σ	RWM proposal step size
$A(t)$	Intermediate bound on $\mathbb{E}\ \mathbf{L}\ _{\mathbb{F}}^2$ feeding u_0
$u_0(t)$	Log of the burn-in chi-squared divergence $\chi^2(\mu_t^0 \ \mu_t)$
R_1	Bound on $\mathbb{E}_{\mu}[R(\mathbf{L})]$
R_2	Bound on $\mathbb{E}_{\mu}[R(\mathbf{L})^2]$
$\hat{R}_2$	$R_1^2 + 2BR_1$
$\hat{R}_4$	Bound on the centered 4th moment of $R(\mathbf{L})$
L_f	Lipschitz constant of the TI score function
$\hat{V}$	Variance bound on $R(\mathbf{L})$ under the finite- K chain
Δ	Maximum τ -grid interval width
$K(t)$	RWM steps per chain at $\tau = t$
$\mathcal{W}_1$	Achieved TV distance at this (Q, ε_*)

Corollary 1. *If our prior satisfies*

$$\tau_{\max} = \Theta(1), \quad \tau_{\min} = \Theta(1), \quad B = \Theta(1),$$

$$\lambda = \Theta(n_1^a), \quad a \leq 1/2,$$

and we standardize the observations $\mathbf{Y}$ such that

$$\sum_{i,j} Y_{ij} \Omega_{ij} = 0, \quad \sum_{i,j} Y_{ij}^2 \Omega_{ij} = \|\mathbf{Y} \circ \mathbf{\Omega}\|_{\mathbb{F}}^2 = \Theta(N),$$

we can achieve η TV distance in

$$QMK = \tilde{O}\left(\frac{N^3 n_1^{2-2a} n_2}{\eta^4}\right)$$

RWM steps.

Corollary 2. *Under the same setting as Corollary 1, we can achieve $n_1 n_2 \eta$ W1 distance in*

$$QMK = \tilde{O}\left(\frac{n_1^{1-2a}}{\eta^4}\right)$$

RWM steps.

We note that the η^4 in the denominator comes entirely from Q . It remains an open question the extent to which Q can be tightened. The desired error tolerance for Corollary 2 comes from the fact that the W1 distance is the infimum of an expected entrywise 1-norm difference, so it makes sense to consider the average error over each entry, yielding a total error tolerance of $n_1 n_2 \eta$. Standardization of observations is a common practice in empirical studies, but to match the scale given by the prior, we scale the entries as in 1.

Table 2: Values of auxiliary quantities

Symbol	Value
σ	$\frac{\lambda}{4n_1\sqrt{n_2}}$
$A(t)$	$\frac{Bn_1n_2}{2} + \left(1 - \frac{2}{Bt+2}\right)^2 \ \mathbf{Y} \circ \mathbf{\Omega}\ _F^2$
$u_0(t)$	$\exp\left(\frac{\sqrt{n_1A(t)}}{\lambda} + \frac{Bn_1}{4\lambda^2}\right) - 1$
R_1	$\ \mathbf{Y} \circ \mathbf{\Omega}\ _F^2 + \frac{NB}{2}$
R_2	$\left(\ \mathbf{Y} \circ \mathbf{\Omega}\ _F + \left(\frac{N(N+2)B^2}{4}\right)^{\frac{1}{4}}\right)^4$
$\tilde{R}_2$	$R_1^2 + 2BR_1$
$\hat{R}_4$	$512B^2R_1^2 + 3076B^4$
L_f	$\frac{N}{2\tau_1^2} + \frac{B\tilde{R}_1}{2}$
$\hat{V}$	$2BR_1 + \sqrt{\hat{R}_4} \varepsilon_\star$
Δ	$\tau_Q(\kappa^{1/Q} - 1)$
$K(t)$	$\geq 2 + 32768 \cdot \frac{Bn_1^2n_2}{C_\ell^2\lambda^2} \exp\left(2 + \frac{(Bt+2)\lambda^2}{8Bn_1}\right)$
	$\left[4(\log \log(u_0(t)/2) - \log \log 4)\right.$
	$\left. + \frac{1}{\log 4} \log\left(\frac{\min\{u_0(t), 8\}}{\varepsilon_\star^2}\right)\right]$
$\mathcal{W}_1$	$\tau_{\max}(\kappa^{1/Q} - 1) + \frac{(\tau_Q - \tau_1)\log^2 \kappa}{16Q^2}$
	$\left(\left(\frac{N}{2} + \frac{\tau_{\max}R_1}{2}\right)^2 + \frac{\tau_{\max}R_1}{6} + \frac{\tau_{\max}^2R_2}{12}\right)$
	$+ \frac{(\tau_Q - \tau_1)^2}{4} \left(L_f \Delta + \sqrt{\tilde{R}_2} \varepsilon_\star\right)$
	$+ \frac{(\tau_Q - \tau_1)^{1.5}}{2} \sqrt{\frac{\hat{V}\Delta}{M}}$

6 Numerical illustration

The contribution of this paper is a feasibility result: a rigorous, high-accuracy sampler for this model exists in polynomial time, at a rate whose constants we do not optimize (Theorems 1 and 2). Accordingly, we do not attempt to validate that rate numerically here; doing so would require running Algorithm 1 for the step counts $K(\tau)$ of Theorem 1, which are very possibly loose and intractable at any scale small enough to plot. The illustrations below instead target two narrower questions: (i) is the degeneracy argument motivating our use of thermodynamic integration visible on a toy problem, and (ii) is the resulting posterior, sampled at practical rather than worst-case step budgets, consistent with the heuristic Gibbs sampler already used in practice. Code and full diagnostics (acceptance rates, split-half Monte Carlo error) are in the supplementary material.

Naïve importance sampling degenerates; thermodynamic integration does not. Estimating the marginal likelihood at a candidate precision $\tau_\star$ by importance-reweighting samples drawn from the untilted prior μ requires importance weights $w_i \propto \exp(-\frac{\tau_\star}{2} R(\mathbf{L}_i))$ that degrade as the tilt moves away from μ . Figure 1 confirms this directly on matrices of size $n \times n$ for $n = 6, \dots, 16$, with N (the number of observed entries) swept from roughly 2 to 200 by scaling n and the observation probability jointly. Panel (A) reports the effective-sample-size fraction ESS/M of the naive estimator, which collapses by roughly two orders of magnitude over this range. Panel (B) reports the

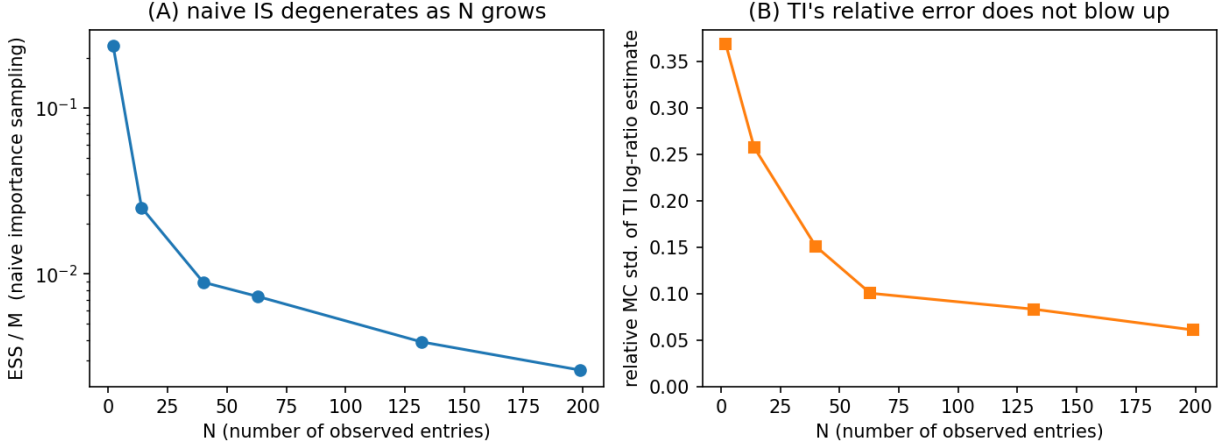


Figure 1: Marginalizing τ on toy $n \times n$ matrices, $n = 6, \dots, 16$. (A) The effective-sample-size fraction of naive importance sampling from the untilted prior collapses as N grows. (B) The relative Monte Carlo error of the thermodynamic-integration estimator over the same range of N stays bounded, at a comparable total sampling budget. Different units in each panel; do not compare on one axis.

relative Monte Carlo standard error of our thermodynamic-integration estimator (the score identity of Appendix B.1, integrated along a path of inner chains that sample only from the correctly tilted measure μ_τ at each grid point), using a comparable total sampling budget; this error stays bounded, and in fact improves, over the same range of N . The two panels use different metrics in different units and should not be read off a shared axis.

Agreement with the heuristic Gibbs sampler, at practical step budgets. We next compare a practical-scale, thermodynamic-integration-based sampler (in the spirit of Algorithm 3, but at step counts tuned for the classical 0.2–0.4 random-walk-Metropolis acceptance regime rather than $K(\tau)$) against Gibbs sampling on $\mathbf{L} \mid \tau$ and $\tau \mid \mathbf{L}$, the heuristic which practitioners already use, and which generally carries no non-asymptotic guarantee. At $n = 12$, on a rank-2 ground truth, with $\lambda = 0.2$, the two samplers agree quantitatively, not just visually: the cross-method gap between their posterior means of $\mathbf{L}$ is smaller than each method’s own split-half Monte Carlo noise (Figure 2). This is evidence that the target distribution our sampler is provably close to is not a pathological artifact of the construction – it is not evidence about the stated convergence rate, which neither chain here is run anywhere near.

Implementation checks. Independently of the above, we verified our random-walk-Metropolis implementation against three closed-form consequences of the model: that μ ’s symmetry under $\mathbf{L} \rightarrow -\mathbf{L}$ forces $\mathbb{E}_\mu[\mathbf{L}] = 0$; the Brascamp–Lieb bound $\text{Var}(L_{ij}) \leq B/2$ implied by μ ’s $2/B$ -strong log-concavity; and the bound $\mathbb{E}_\mu[R(\mathbf{L})] \leq R_1$ of Lemma 2. All three hold empirically; details are in the supplementary code.

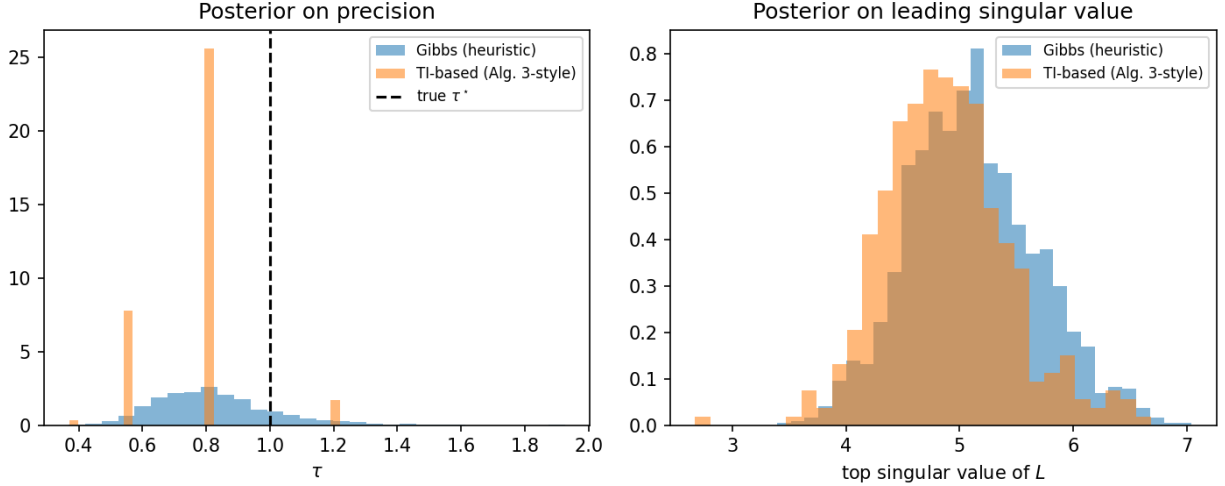


Figure 2: Posterior on the noise precision τ and on the leading singular value of $\mathbf{L}$: Gibbs sampling (heuristic, no guarantee) versus our thermodynamic-integration-based sampler, at matched practical step budgets. The cross-method gap is within each method’s own split-half Monte Carlo noise.

References

- Abadie, A., Diamond, A., and Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of california’s tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505.
- Abrahamsen, T. and Hobert, J. P. (2017). Convergence analysis of block Gibbs samplers for Bayesian linear mixed models with *pgt*; *N. Bernoulli*, 23(1):459 – 478.
- Agarwal, A., Dahleh, M., Shah, D., and Shen, D. (2021). Causal matrix completion. *arXiv preprint arXiv:2109.15154*.
- Andrieu, C., Lee, A., Power, S., and Wang, A. Q. (2024). Explicit convergence bounds for Metropolis Markov chains: isoperimetry, spectral gaps and profiles. *Annals of Applied Probability*, 34(4):4022–4071.
- Athey, S., Bayati, M., Doudchenko, N., Imbens, G., and Khosravi, K. (2021). Matrix completion methods for causal panel data models. *Journal of the American Statistical Association*, 116(536):1716–1730.
- Bai, J. and Ng, S. (2021). Matrix completion, counterfactuals, and factor analysis of missing data. *Journal of the American Statistical Association*, 116(536):1746–1763.
- Candès, E. and Tao, T. (2010). The power of convex relaxation: Near-optimal matrix completion. *IEEE Transactions on Information Theory*, 56:2053–2080.
- Chen, Y., Dwivedi, R., Wainwright, M. J., and Yu, B. (2020a). Fast mixing of metropolized hamiltonian monte carlo: benefits of multi-step gradients. *J. Mach. Learn. Res.*, 21(1).
- Chen, Y., Fan, J., Ma, C., and Yan, Y. (2020b). Bridging convex and nonconvex optimization in robust pca: Noise, outliers, and missing data. *Annals of statistics*, 49 5:2948–2971.

- Chewi, S. (2024). *Log-Concave Sampling*.
- Cui, T. and Gorodetsky, A. (2024). Low-rank bayesian matrix completion via geodesic hamiltonian monte carlo on stiefel manifolds. *ArXiv*.
- Durmus, A. and Moulines, É. (2017). Nonasymptotic convergence analysis for the unadjusted Langevin algorithm. *The Annals of Applied Probability*, 27(3):1551 – 1587.
- Durmus, A., Moulines, E., and Pereyra, M. (2022). A proximal markov chain monte carlo method for bayesian inference in imaging inverse problems: When langevin meets moreau. *SIAM Review*, 64(4):991–1028.
- Dwivedi, R., Chen, Y., Wainwright, M. J., and Yu, B. (2019). Log-concave sampling: Metropolis-hastings algorithms are fast. *Journal of Machine Learning Research*, 20(183):1–42.
- Hobert, J. P. and Geyer, C. J. (1998). Geometric ergodicity of gibbs and block gibbs samplers for a hierarchical random effects model. *Journal of Multivariate Analysis*, 67(2):414–430.
- Jones, G. L. and Hobert, J. P. (2004). Sufficient burn-in for Gibbs samplers for a hierarchical random effects model. *The Annals of Statistics*, 32(2):784 – 817.
- Keshavan, R. H. and Montanari, A. (2010). Regularization for matrix completion. *2010 IEEE International Symposium on Information Theory*, pages 1503–1507.
- Keshavan, R. H., Montanari, A., and Oh, S. (2010). Matrix completion from noisy entries. *Journal of Machine Learning Research*, pages 2057–2078.
- Koren, Y., Bell, R., and Volinsky, C. (2009). Matrix factorization techniques for recommender systems. *Computer*, 42(8):30–37.
- Lee, Y. T., Shen, R., and Tian, K. (2020). Logsmooth gradient concentration and tighter runtimes for metropolized hamiltonian monte carlo. In Abernethy, J. and Agarwal, S., editors, *Proceedings of Thirty Third Conference on Learning Theory*, volume 125 of *Proceedings of Machine Learning Research*, pages 2565–2597. PMLR.
- Mou, W., Flammarion, N., Wainwright, M. J., and Bartlett, P. L. (2022). An efficient sampling algorithm for non-smooth composite potentials. *Journal of Machine Learning Research*, 23(233):1–50.
- Negahban, S. and Wainwright, M. (2010). Restricted strong convexity and weighted matrix completion: Optimal bounds with noise. *J. Mach. Learn. Res.*, 13:1665–1697.
- Román, J. C. and Hobert, J. P. (2012). Convergence analysis of the Gibbs sampler for Bayesian general linear mixed models with improper priors. *The Annals of Statistics*, 40(6):2823 – 2849.
- Rosenthal, J. S. (1995). Rates of convergence for gibbs sampling for variance component models. *The Annals of Statistics*, 23(3):740–761.
- Salim, A., Kovalev, D., and Richtarik, P. (2019). Stochastic proximal langevin algorithm: Potential splitting and nonasymptotic rates. In Wallach, H., Larochelle, H., Beygelzimer, A., d'Alché-Buc, F., Fox, E., and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

- Salim, A. and Richtarik, P. (2020). Primal dual interpretation of the proximal stochastic gradient langevin algorithm. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*, volume 33, pages 3786–3796. Curran Associates, Inc.
- Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society. Series B (Methodological)*, 58(1):267–288.

Appendix

A Chi-squared mixing bounds for high-accuracy samplers

A.1 Setup

We want to sample from the following measure on $\mathbb{R}^d$:

$$\pi(dx) \propto \exp(-V(x)) dx, \quad V(x) = F(x) + G(x),$$

where F is m -strongly convex and L_F -smooth and G is convex but nonsmooth and not everywhere differentiable, but it is L_G -Lipschitz, which implies that its minimal subgradient satisfies $\|\nabla^0 G\| \leq L_G$. This problem is more difficult than the classic smooth-and-strongly-convex setting. We will sample using a Metropolis-adjusted Markov chain P targeting π , and we want a bound of the form

$$\chi^2(\pi_0 P^K \|\pi) \leq \varepsilon_{\text{mix}}, \quad K = \text{poly}(d, \log(1/\varepsilon_{\text{mix}})).$$

We will lean heavily on [Andrieu et al. \(2024\)](#) and begin by giving an overview of the terminology used in that paper.

- For a π -measurable set A , the **conductance**

$$\Phi(A) = \frac{1}{\pi(A)} \int_A P(\mathbf{L}, A^c) \pi(d\mathbf{L})$$

measures how much probability mass flows between A and A^c per step.

- For a set A with $\pi(A) = p$, its **boundary measure**

$$\pi^+(A) := \liminf_{r \rightarrow 0} \left\{ \frac{\pi(B_r(A)) - \pi(A)}{r} \right\}$$

measures how much π -mass sits at the edge of A .

- The **isoperimetric profile** $I_\pi(p) := \inf\{\pi^+(A) : \pi(A) = p\}$ measures the smallest perimeter (boundary measure) of a set of probability p . Intuitively, if I_π is large, the boundaries are large, so each set of a given sides has many places to exit and the Markov chain will not be stuck one one side of any boundary.
- An **isoperimetric minorant** $\tilde{I}_\pi \leq I_\pi$ is any lower bound on I_π .
- A minorant is **regular** if it is symmetric about $p = 1/2$, continuous, and increasing on $(0, 1/2]$.
- The Markov kernel P is (δ, ε) -**close-coupling** if any two starting points within Euclidean distance δ of each other have next-step distributions within $1 - \varepsilon$ TV distance of each other.
- The **spectral gap**

$$\gamma_P := 1 - \|P\|_{L_0^2}, \quad \|P\|_{L_0^2} = \sup_{\|f\|_2=1, \int f d\pi=0} \|Pf\|_2,$$

controls geometric decay of chi-squared divergence:

$$\chi^2(\pi_0 P^k \|\pi) \leq (1 - \gamma_P)^{2k} \chi^2(\pi_0 \|\pi).$$

A.2 The general recipe

The main result of [Andrieu et al. \(2024\)](#) (Theorem 18) states that if

- π has regular concave isoperimetric minorant $\tilde{I}_\pi$,
- P is (δ, ε) -close-coupling and π -reversible, and
- $u_0 := \chi^2(\pi_0 \| \pi)$ is finite,

we have $\chi^2(\pi_0 P^k \| \pi) \leq \varepsilon_{\text{mix}}$ for $\varepsilon_{\text{mix}} \in (0, 8)$ after

$$\begin{aligned} K \geq & 2 + \frac{64}{\varepsilon^2} \max \left\{ \log \left(\frac{u_0 v_*}{4} \right), 0 \right\} && \text{(term 1: burn-in)} \\ & + \frac{256}{\varepsilon^2 \delta^2} \int_{\max\{2/u_0, 1/4\}, v_*/2}^{1/4} \frac{\xi}{\tilde{I}_\pi(\xi)^2} d\xi && \text{(term 2: bulk mixing)} \\ & + \frac{16}{\varepsilon^2} \max \left\{ 1, \frac{1}{4\delta^2 \tilde{I}_\pi(1/4)^2} \right\} \max \left\{ \log \left(\frac{\min\{u_0, 8\}}{\varepsilon_{\text{mix}}} \right), 0 \right\} && \text{(term 3: fine convergence)} \end{aligned}$$

where v_* is a threshold scale solving

$$v_* := \min \left\{ \frac{1}{2}, \max \left\{ 0, \sup \left\{ v > 0 : \tilde{I}_\pi \left(\frac{v}{2} \right) \geq \frac{v}{\delta} \right\} \right\} \right\}.$$

This is exactly the scaling we need; all that remains is to find $\delta, \varepsilon, \tilde{I}_\pi$ for our specific setting. To do this, we use another result from the paper: for any Metropolis kernel with proposal Q , we have

$$\|P(x, \cdot) - P(y, \cdot)\|_{\text{TV}} \leq \|Q(x, \cdot) - Q(y, \cdot)\|_{\text{TV}} + 1 - \alpha_0, \quad \alpha_0 := \inf_x \alpha(x),$$

where $\alpha(x)$ is the average acceptance probability at x . This splits the close-coupling problem into two independent pieces:

1. How stable is the proposal as we move the current point?
2. How low can the acceptance probability get?

We will solve each separately and add the bounds.

A.3 The target-side ingredient

For any m -strongly convex potential, Caffarelli's contraction theorem gives a 1-Lipschitz map pushing $\mathcal{N}(0, m^{-1}\mathbf{I})$ onto π , so π inherits the Gaussian isoperimetric profile, scaled by $\sqrt{m}$ ([Andrieu et al. \(2024\)](#), Lemma 27):

$$I_\pi(\xi) \geq \tilde{I}_\pi(\xi) := C_\ell \xi \sqrt{m \log(1/\xi)}, \quad \xi \in (0, 1/2), \quad C_\ell = 0.958357.$$

This is dimension-free!

A.4 The algorithm-side ingredients

Andrieu et al.’s acceptance-probability lemma (their Lemma 39) requires a one-sided growth bound

$$V(\mathbf{x} + \mathbf{h}) - V(\mathbf{x}) - \langle \nabla V(\mathbf{x}), \mathbf{h} \rangle \leq \psi(\|\mathbf{h}\|),$$

for some nondecreasing ψ . We therefore have

$$\begin{aligned} & V(\mathbf{x} + \mathbf{h}) - V(\mathbf{x}) - \langle \nabla^0 V(\mathbf{x}), \mathbf{h} \rangle \\ &= F(\mathbf{x} + \mathbf{h}) - F(\mathbf{x}) - \langle \nabla F(\mathbf{x}), \mathbf{h} \rangle + G(\mathbf{x} + \mathbf{h}) - G(\mathbf{x}) - \langle \nabla^0 G(\mathbf{x}), \mathbf{h} \rangle \\ &\leq \frac{L_F}{2} \|\mathbf{h}\|^2 + 2L_G \|\mathbf{h}\| \\ &=: \psi(\|\mathbf{h}\|). \end{aligned}$$

For random-walk Metropolis, the proposal is $\mathcal{N}(\mathbf{x}, \sigma^2 \mathbf{I})$. By the Gaussian KL identity and Pinsker’s inequality,

$$\|Q_x - Q_y\|_{\text{TV}} \leq \sqrt{\frac{1}{2} \text{KL}(Q_x \| Q_y)} = \frac{\|x - y\|}{2\sigma}.$$

Andrieu et al. (2024)’s symmetrization argument (Lemma 39) gives, for any growth bound ψ ,

$$\alpha_0 \geq \frac{1}{2} \exp(-\mathbb{E}_{\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})}[\psi(\sigma \|\mathbf{z}\|)]).$$

Plugging in our ψ and choosing $\sigma = \varsigma / L_G \sqrt{d}$ for a tuning constant $\varsigma > 0$ makes both terms of $\mathbb{E}[\psi(\sigma \|\mathbf{z}\|)]$ bounded, dimension-free constants:

$$\alpha_0 \geq \frac{1}{2} \exp\left(-2\varsigma - \frac{L_F \varsigma^2}{2L_G^2}\right).$$

A.5 Assembling the mixing bound

We will evaluate the integral from term 2 explicitly. We find that the equation $\tilde{I}_\pi(v/2) = v/\delta$ is solved by

$$v_* = 2 \exp\left(-\frac{4}{C_\ell^2 m \delta^2}\right),$$

which will be squared-exponentially small if δ is small. If $u_0 < 4/v_*$, which will always be the case in our applications because $4/v_*$ is exponentially large, the first term is zero. In that case, the integral in the second term will come out to

$$\int_{2/u_0}^{1/4} \frac{\xi}{\tilde{I}_\pi(\xi)^2} d\xi = \frac{1}{C_\ell^2 m} \int_{2/u_0}^{1/4} \frac{1}{\xi \log(1/\xi)} d\xi = \frac{\log \log(u_0/2) - \log \log(4)}{C_\ell^2 m}.$$

Our bound is now

$$\begin{aligned} K &\geq 2 + \frac{256}{\varepsilon^2 \delta^2} \cdot \frac{\log \log(u_0/2) - \log \log(4)}{C_\ell^2 m} + \frac{64}{\varepsilon^2} \cdot \frac{1}{C_\ell^2 m \delta^2 \log(4)} \log\left(\frac{\min\{u_0, 8\}}{\varepsilon_{\text{mix}}}\right) \\ &= 2 + \frac{64}{C_\ell^2 m \varepsilon^2 \delta^2} \left[4(\log \log(u_0/2) - \log \log(4)) + \frac{1}{\log(4)} \log\left(\frac{\min\{u_0, 8\}}{\varepsilon_{\text{mix}}}\right)\right]. \end{aligned}$$

Now it remains to optimize our ε and δ . Combining the earlier results, we see that

$$\|P(x, \cdot) - P(y, \cdot)\|_{\text{TV}} \leq \|Q(x, \cdot) - Q(y, \cdot)\|_{\text{TV}} + 1 - \alpha_0 \leq \frac{\|x - y\|}{2\sigma} + 1 - \alpha_0.$$

To turn this into a (δ, ε) -close-coupling, we need a bound δ on the Euclidean distance and a bound $1 - \varepsilon$ on the TV distance. We choose $\delta = \alpha_0 \sigma$ so that the first term is at most half of what α_0 allows. The resulting right side of the inequality above tells us to set $\varepsilon = \alpha_0/2$. Not knowing α_0 , we substitute the lower bound, and we get δ and ε as functions solely of ς :

$$\delta = \frac{\varsigma}{2L_G\sqrt{d}} \exp\left(-2\varsigma - \frac{L_F\varsigma^2}{2L_G^2}\right), \quad \varepsilon = \frac{1}{4} \exp\left(-2\varsigma - \frac{L_F\varsigma^2}{2L_G^2}\right).$$

Looking at our formula for the necessary K , we see that to minimize it, we must maximize $\varepsilon\delta$. Neglecting the quadratic term in the exponent, which vanishes quickly, we can see from first-order optimality that an approximate optimum occurs at $\varsigma = 1/4$, which yields

$$\varepsilon\delta = \frac{1}{32L_G\sqrt{d}} \exp\left(-1 - \frac{L_F}{16L_G^2}\right).$$

Our final bound is

$$K \geq 2 + 65536 \cdot \frac{L_G^2 d}{C_\ell^2 m} \exp\left(2 + \frac{L_F}{8L_G^2}\right) \left[4(\log \log(u_0/2) - \log \log(4)) + \frac{1}{\log(4)} \log\left(\frac{\min\{u_0, 8\}}{\varepsilon_{\text{mix}}}\right)\right].$$

The $\exp(L_F/8L_G^2)$ factor is the cost of using a growth bound derived from [Andrieu et al. \(2024\)](#) rather than a proximal method tailored to the nuclear norm. However, this is still necessary to prevent exponential dependence on dimension.

A.6 A bound on the initial chi-squared divergence

The previous sections' results hinge on u_0 , which is often difficult to compute or bound. However, in the applications we are considering, an obvious initialization measure exists with tractable bounds on u_0 .

Lemma 1. *Let*

$$\pi(d\mathbf{L}) \propto \exp\left(-\sum_{i,j} \phi_{ij}(L_{ij}) - \frac{1}{\lambda} \|\mathbf{L}\|_*\right) d\mathbf{L},$$

where each $\phi_{ij} : \mathbb{R} \rightarrow \mathbb{R}$ is m -strongly convex. Define the product measure

$$\pi^0(d\mathbf{L}) \propto \exp\left(-\sum_{i,j} \phi_{ij}(L_{ij})\right) d\mathbf{L}.$$

Then

$$\chi^2(\pi^0 \| \pi) \leq \exp\left(\frac{\sqrt{r}}{\lambda} \sqrt{\frac{d}{m} + \|\mathbb{E}_{\pi^0}[\mathbf{L}]\|_{\mathbb{F}}^2} + \frac{r}{2m\lambda^2}\right) - 1,$$

where $d = n_1 n_2$ and $r = \min\{n_1, n_2\}$.

Proof. Let

$$Z_\pi = \int \exp\left(-\sum_{i,j} \phi_{ij}(L_{ij}) - \frac{1}{\lambda} \|\mathbf{L}\|_*\right) d\mathbf{L}, \quad Z_{\pi^0} = \int \exp\left(-\sum_{i,j} \phi_{ij}(L_{ij})\right) d\mathbf{L}.$$

Since the nuclear norm is nonnegative, $Z_\pi \leq Z_{\pi^0}$. Moreover,

$$\frac{d\pi^0}{d\pi}(\mathbf{L}) = \frac{Z_\pi}{Z_{\pi^0}} \exp\left(\frac{1}{\lambda} \|\mathbf{L}\|_*\right),$$

so

$$\chi^2(\pi^0 \|\pi) = \int \left(\frac{d\pi^0}{d\pi}\right)^2 d\pi - 1 = \frac{Z_\pi}{Z_{\pi^0}} \mathbb{E}_{\pi^0} \left[\exp\left(\frac{1}{\lambda} \|\mathbf{L}\|_*\right) \right] - 1 \leq \mathbb{E}_{\pi^0} \left[\exp\left(\frac{1}{\lambda} \|\mathbf{L}\|_*\right) \right] - 1.$$

Using $\|\mathbf{L}\|_* \leq \sqrt{r} \|\mathbf{L}\|_F$, we obtain

$$\chi^2(\pi^0 \|\pi) \leq \mathbb{E}_{\pi^0} \left[\exp\left(\frac{\sqrt{r}}{\lambda} \|\mathbf{L}\|_F\right) \right] - 1.$$

Since each ϕ_{ij} is m -strongly convex, the product measure μ is m -strongly log-concave. The function $f(\mathbf{L}) = \|\mathbf{L}\|_F$ is 1-Lipschitz with respect to the Frobenius norm. The Gaussian concentration inequality for strongly log-concave measures gives

$$\mathbb{E}_{\pi^0} \left[\exp\left(\frac{\sqrt{r}}{\lambda} f\right) \right] \leq \exp\left(\frac{\sqrt{r}}{\lambda} \mathbb{E}_{\pi^0}[f] + \frac{r}{2m\lambda^2}\right).$$

By the Brascamp–Lieb inequality,

$$\mathbb{V}[L_{ij}] \leq \frac{1}{m},$$

so

$$\mathbb{E}_{\pi^0}[\|\mathbf{L}\|_F^2] \leq \frac{d}{m} + \|\mathbb{E}_{\pi^0}[\mathbf{L}]\|_F^2.$$

By Jensen's inequality,

$$\mathbb{E}_{\pi^0}[\|\mathbf{L}\|_F] \leq \sqrt{\frac{d}{m} + \|\mathbb{E}_{\pi^0}[\mathbf{L}]\|_F^2}.$$

Therefore,

$$\mathbb{E}_{\pi^0} \left[\exp\left(\frac{\sqrt{r}}{\lambda} f\right) \right] \leq \exp\left(\frac{\sqrt{r}}{\lambda} \sqrt{\frac{d}{m} + \|\mathbb{E}_{\pi^0} \mathbf{L}\|_F^2} + \frac{r}{2m\lambda^2}\right),$$

and our desired result follows. $\square$

B Some moment bounds and transport inequalities

Before we introduce the following lemmas, it will be useful to define the following for a measure ρ :

- $Z_\rho(s) := \mathbb{E}_\rho \left[\exp\left(-\frac{s}{2} R(\mathbf{L})\right) \right]$ is the scaling constant of ρ_s .
- $\Lambda_\rho(s) := \log(s^{N/2} Z_\rho(s))$ is the part of the log-posterior which depends on s .
- $\hat{\Lambda}_\rho(I) = \log \int_I t^{N/2} Z_\rho(t) dt$ is the log-averaged version of $\Lambda_\rho(s)$ over I .
- $H_\rho(s) = \exp\left(\frac{sN}{2}\right) Z_\rho(e^s)$ is τ on a logarithmic scale, so $s_q := \log \tau_q$ is the midpoint of the interval $\log I_q$, which has width $\Delta_s := (\log \kappa)/Q$.

Lemma 2. *We have $\mathbb{E}_\mu[R(\mathbf{L})] \leq R_1$.*

Proof. Because μ is $2/B$ -strongly log-concave, the Brascamp-Lieb inequality gives

$$\text{Cov}_\mu(\mathbf{L}) \preceq \frac{B}{2} \mathbf{I}.$$

Taking traces,

$$\mathbb{E}_\mu[\|\mathbf{L} \circ \boldsymbol{\Omega}\|_{\text{F}}^2] \leq \frac{NB}{2}.$$

Because $\mathbf{L}$ has mean zero under μ and $\mathbf{Y} \circ \boldsymbol{\Omega}$ is deterministic, we have

$$\mathbb{E}_\mu[\langle \mathbf{L} \circ \boldsymbol{\Omega}, \mathbf{Y} \circ \boldsymbol{\Omega} \rangle_{\text{F}}] = 0,$$

so

$$\mathbb{E}_\mu[\|(\mathbf{L} - \mathbf{Y}) \circ \boldsymbol{\Omega}\|_{\text{F}}^2] = \|\mathbf{Y} \circ \boldsymbol{\Omega}\|_{\text{F}}^2 + \mathbb{E}_\mu[\|\mathbf{L} \circ \boldsymbol{\Omega}\|_{\text{F}}^2] \leq \|\mathbf{Y} \circ \boldsymbol{\Omega}\|_{\text{F}}^2 + \frac{NB}{2}.$$

□

Lemma 3. *We have $\mathbb{E}_\mu[R^2(\mathbf{L})] \leq R_2$.*

Proof. Because μ is $2/B$ -strongly log-concave, by Brascamp-Lieb, its even moments are no greater than those of Gaussian measure over $\mathbb{R}^{n_1 \times n_2}$ where each coordinate is i.i.d. and $2/B$ -strongly log-concave, so we can use Gaussian moments to see that

$$\mathbb{E}_\mu[\|\mathbf{L} \circ \boldsymbol{\Omega}\|_{\text{F}}^4] \leq \frac{N(N+2)B^2}{4}.$$

Using Minkowski in L^4 ,

$$\begin{aligned} \mathbb{E}_\mu[\|(\mathbf{L} - \mathbf{Y}) \circ \boldsymbol{\Omega}\|_{\text{F}}^4] &\leq \left(\|\mathbf{Y} \circ \boldsymbol{\Omega}\|_{\text{F}} + \left(\mathbb{E}_\mu[\|\mathbf{L} \circ \boldsymbol{\Omega}\|_{\text{F}}^4] \right)^{\frac{1}{4}} \right)^4 \\ &\leq \left(\|\mathbf{Y} \circ \boldsymbol{\Omega}\|_{\text{F}} + \left(\frac{N(N+2)B^2}{4} \right)^{\frac{1}{4}} \right)^4. \end{aligned}$$

□

Lemma 4. *We have*

$$W_1(\nu_\star, \nu_\star^Q) \leq \tau_{\max}(\kappa^{1/Q} - 1) + \frac{(\tau_Q - \tau_1) \log^2 \kappa}{16Q^2} \left(\left(\frac{N}{2} + \frac{\tau_{\max} R_1}{2} \right)^2 + \frac{\tau_{\max} R_1}{6} + \frac{\tau_{\max}^2 R_2}{12} \right).$$

Proof. Letting $\kappa = \tau_{\max}/\tau_{\min}$, we have $\tau_q = \tau_{\min} \kappa^{(q-1)/2/Q}$. Let

$$I_q = [\tau_{\min} \kappa^{(q-1)/Q}, \tau_{\min} \kappa^{q/Q}]$$

and define the categorical distribution $\hat{\nu}_\star^Q$ such that $\hat{\nu}_\star^Q(\tau_q) = \nu_\star(I_q)$. By the triangle inequality,

$$W_1(\nu_\star, \nu_\star^Q) \leq W_1(\nu_\star, \hat{\nu}_\star^Q) + W_1(\hat{\nu}_\star^Q, \nu_\star^Q).$$

Using the simple coupling from $\nu_\star$ to $\hat{\nu}_\star^Q$ that maps a sample $\tau \in I_q$ to τ_q , we can see

$$W_1(\nu_\star, \hat{\nu}_\star^Q) \leq \tau_{\max}(\kappa^{1/Q} - 1).$$

If we let $\psi = \log H_\mu$ and $U \sim \text{Unif}[-\Delta_s/2, \Delta_s/2]$, we have

$$\widehat{\Lambda}_\mu(I_q) = \log \left(\int_{s_q - \Delta_s/2}^{s_q + \Delta_s/2} H_\mu(s) ds \right) = \Lambda_\mu(\tau_q) + \log \Delta_s + \log \mathbb{E}_U[\exp(\phi_q(U))]$$

for $\phi_q(u) := \psi(s_q + u) - \psi(s_q)$. By the mean value theorem,

$$|\phi_q(u)| \leq \|\psi'\|_\infty |u| \leq \|\psi'\|_\infty \frac{\Delta_s}{2}.$$

By Hoeffding's lemma, we now have

$$\log \mathbb{E}_U[\exp(\pm \phi_q(U))] \leq \pm \mathbb{E}_U[\phi_q(U)] + \frac{\|\psi'\|_\infty^2 \Delta_s^2}{8}.$$

Using the midpoint rule as an error bound,

$$|\mathbb{E}_U[\phi_q(U)]| = \frac{1}{\Delta_s} \left| \int_{-\Delta_s/2}^{\Delta_s/2} (\psi(s_q + u) - \psi(s_q)) du \right| \leq \frac{\|\psi''\|_\infty \Delta_s^2}{24},$$

so applying in the positive and negative directions, we get

$$|\log \mathbb{E}_U[\exp(\phi_q(U))]| \leq \frac{\|\psi''\|_\infty \Delta_s^2}{24} + \frac{\|\psi'\|_\infty^2 \Delta_s^2}{8}.$$

Since e^s is increasing in s and R is nonnegative and does not depend on s , the absolute values of both derivative bounds are increasing in s , so

$$\begin{aligned} \|\psi'\|_\infty &\leq \frac{N}{2} + \frac{\tau_{\max}}{2} \mathbb{E}_\mu[R] \leq \frac{N}{2} + \frac{\tau_{\max}}{2} R_1 \text{ and} \\ \|\psi''\|_\infty &\leq \frac{\tau_{\max}}{2} \mathbb{E}_\mu[R] + \frac{\tau_{\max}^2}{4} \mathbb{V}_\mu[R] \leq \frac{\tau_{\max}}{2} R_1 + \frac{\tau_{\max}^2}{4} R_2. \end{aligned}$$

On a bounded set, we have

$$W_1(\widehat{\nu}_\star^Q, \nu_\star^Q) \leq (\tau_Q - \tau_1) \|\widehat{\nu}_\star^Q - \nu_\star^Q\|_{\text{TV}} \leq \frac{\tau_Q - \tau_1}{2} \max_q |\log \mathbb{E}_U[\exp(\phi_q(U))]|,$$

by shift-invariance of the softmax function, implying our desired result. $\square$

B.1 The thermodynamic integration step

We first compute the score and curvature of Λ_μ , which gives both a derivative bound and a direct proof of the monotonicity of $\mathbb{E}_{\nu_t}[R(\mathbf{L})]$ in t , used informally in the proof of Lemma 4. Because

$$\mathbb{E}_{\mu_s}[R] = \frac{\mathbb{E}_\mu[R \exp(-sR/2)]}{Z_\mu(s)}, \quad \frac{d\mu_s}{d\mu} = \frac{\exp(-sR/2)}{Z_\mu(s)} \Rightarrow \frac{Z'_\mu(s)}{Z_\mu(s)} = -\frac{1}{2} \mathbb{E}_{\mu_s}[R],$$

we have

$$\frac{d}{ds} \mathbb{E}_{\mu_s}[R] = -\frac{1}{2} \mathbb{E}_{\mu_s}[R^2] - \mathbb{E}_{\mu_s}[R] \frac{Z'_\mu(s)}{Z_\mu(s)} = -\frac{1}{2} \mathbb{E}_{\mu_s}[R^2] + \frac{1}{2} \mathbb{E}_{\mu_s}[R]^2 = -\frac{1}{2} \mathbb{V}_{\mu_s}[R].$$

Differentiating $\Lambda_\mu(s) = \frac{N}{2} \log s + \log Z_\mu(s)$ and using $\frac{d}{ds} \log Z_\mu(s) = -\frac{1}{2} \mathbb{E}_{\mu_s}[R]$, we obtain

$$\Lambda'_\mu(s) = \frac{N}{2s} - \frac{1}{2} \mathbb{E}_{\mu_s}[R(L)], \quad \Lambda''_\mu(s) = -\frac{N}{2s^2} + \frac{1}{4} \mathbb{V}_{\mu_s}[R(L)].$$

Step 4 of the proof below also needs a uniform bound on the fourth central moment of R under the tilted family; we record it now so it is available where needed.

Lemma 5 (Centered fourth moment of R , uniform over the tilted family). *For all $s \geq 0$,*

$$\mathbb{E}_{\mu_s}[(R(\mathbf{L}) - \mathbb{E}_{\mu_s}[R(\mathbf{L})])^4] \leq \hat{R}_4.$$

Proof. Let $f(\mathbf{L}) := \sqrt{R(\mathbf{L})}$. Since Ω has entries in $\{0, 1\}$,

$$|f(\mathbf{L}) - f(\mathbf{L}')| \leq \|(\mathbf{L} - \mathbf{L}') \circ \Omega\|_F \leq \|\mathbf{L} - \mathbf{L}'\|_F,$$

so f is 1-Lipschitz in Frobenius norm. μ_s is $2/B$ -strongly log-concave for every s , so by Caffarelli's contraction theorem (Appendix A.3), there is a 1-Lipschitz map pushing $\mathcal{N}(0, \frac{B}{2}\mathbf{I})$ onto μ_s ; composing with the 1-Lipschitz f and applying Gaussian concentration gives

$$\mu_s(|f - \mathbb{E}_{\mu_s} f| > t) \leq 2 \exp(-t^2/B) \forall t > 0.$$

(This is the same concentration fact used in the proof of Lemma 1, applied here to f instead of $\|\mathbf{L}\|_F$.) **Moments of the centered part.** Let $g := f - \bar{f}$ where $\bar{f} := \mathbb{E}_{\mu_s}[f]$. For even $p \geq 2$, tail-integrating the bound above (substitute $u = t^2/B$),

$$\mathbb{E}[|g|^p] = \int_0^\infty p t^{p-1} \mu_s(|g| > t) dt \leq 2p \int_0^\infty t^{p-1} e^{-t^2/B} dt = p\Gamma(p/2)B^{p/2}.$$

In particular, $\mathbb{E}[g^4] \leq 4\Gamma(2)B^2 = 4B^2$ and $\mathbb{E}[g^8] \leq 8\Gamma(4)B^4 = 48B^4$. Directly by Poincaré (as in Step 2 below, using $\|\nabla f\| = 1$ a.e.),

$$\mathbb{E}[g^2] = \mathbb{V}_{\mu_s}[f] \leq \frac{B}{2} \mathbb{E}_{\mu_s}[\|\nabla f\|^2] \leq \frac{B}{2}.$$

From moments of f to a centered moment of R . Write $R = f^2 = \bar{f}^2 + 2\bar{f}g + g^2$, so with $\bar{R} := \mathbb{E}_{\mu_s}[R] = \bar{f}^2 + \mathbb{E}[g^2]$,

$$R - \bar{R} = 2\bar{f}g + (g^2 - \mathbb{E}[g^2]).$$

Using $(a + b)^4 \leq 8(a^4 + b^4)$ twice, we have

$$(R - \bar{R})^4 \leq 8(16\bar{f}^4 g^4 + (g^2 - \mathbb{E}[g^2])^4), \quad (g^2 - \mathbb{E}[g^2])^4 \leq 8(g^8 + (\mathbb{E}[g^2])^4).$$

Taking expectations,

$$\mathbb{E}[(R - \bar{R})^4] \leq 128\bar{f}^4 \mathbb{E}[g^4] + 64(\mathbb{E}[g^8] + (\mathbb{E}[g^2])^4) = 512\bar{f}^4 B^2 + 3076B^4.$$

Finally, $\bar{f}^2 \leq \mathbb{E}_{\mu_s}[f^2] = \mathbb{E}_{\mu_s}[R] \leq R_1$ by Jensen, monotonicity, and Lemma 2, so $\bar{f}^4 \leq R_1^2$, giving the claim. $\square$

Lemma 6 (Thermodynamic-integration bound). *We have*

$$\mathbb{E}\left[W_1(\nu_\star^Q, \nu_\star^{Q, \text{TI}})\right] \leq \frac{(\tau_Q - \tau_1)^2}{4} \left(L_f \Delta + \sqrt{\tilde{R}_2} \varepsilon_\star\right) + \frac{\tau_Q - \tau_1}{2} \sqrt{\frac{\hat{V} \Delta (\tau_Q - \tau_1)}{M}}.$$

Proof. **Step 0 (reduce to a sup-norm comparison).** As in the proof of Lemma 4, shift-invariance of the softmax gives

$$\nu_\star^Q = \text{softmax}(\{\tilde{\Lambda}_\mu(\tau_1), \dots, \tilde{\Lambda}_\mu(\tau_Q)\})$$

for

$$\tilde{\Lambda}_\mu(\tau_q) := \Lambda_\mu(\tau_q) - \Lambda_\mu(\tau_1) = \int_{\tau_1}^{\tau_q} f(s) ds, \quad f(s) := \Lambda'_\mu(s) = \frac{N}{2s} - \frac{1}{2} \mathbb{E}_{\mu_s}[R(\mathbf{L})].$$

Because the softmax function is $\frac{1}{2}$ -Lipschitz from the infinity norm to TV distance, we can use the TV bound for W1 distance on a bounded interval to say that

$$W_1(\nu_\star^Q, \nu_\star^{Q, \text{TI}}) \leq (\tau_Q - \tau_1) \|\nu_\star^Q - \nu_\star^{Q, \text{TI}}\|_{\text{TV}} \leq \frac{\tau_Q - \tau_1}{2} \max_q |\tilde{\Lambda}_\mu(\tau_q) - \hat{\Lambda}_q|.$$

Step 1 (exact three-way split). Fix q . Writing $f(\tau_i)(\tau_{i+1} - \tau_i) = (\frac{N}{2\tau_i} - \frac{1}{2}\mathbb{E}_{\mu_{\tau_i}}[R])(\tau_{i+1} - \tau_i)$, we have

$$\begin{aligned} \tilde{\Lambda}_\mu(\tau_q) - \hat{\Lambda}_q &= \left(\int_{\tau_1}^{\tau_q} f(s) ds - \sum_{i < q} f(\tau_i)(\tau_{i+1} - \tau_i) \right) && \text{(term A: quadrature)} \\ &+ \frac{1}{2} \sum_{i < q} (\mathbb{E}_{\mu_{\tau_i}^{K(\tau_i)}}[R] - \mathbb{E}_{\mu_{\tau_i}}[R])(\tau_{i+1} - \tau_i) && \text{(term B: chain bias)} \\ &+ \frac{1}{2} \sum_{i < q} (\hat{R}_M(\tau_i) - \mathbb{E}_{\mu_{\tau_i}^{K(\tau_i)}}[R])(\tau_{i+1} - \tau_i). && \text{(term C: MC noise)} \end{aligned}$$

Terms A and B are differences of fixed expectations and a deterministic quadrature error respectively, so only term C is random, which matters in Step 4.

Step 2 (term A). We already saw $f'(s) = -\frac{N}{2s^2} + \frac{1}{4}\mathbb{V}_{\mu_s}[R]$. By the Brascamp–Lieb/Poincaré inequality under μ_s 's strong-convexity constant $2/B$ applied to R itself, and using $\|\nabla R\|^2 = 4R$, we have

$$\mathbb{V}_{\mu_s}[R] \leq \frac{B}{2} \mathbb{E}_{\mu_s}[\|\nabla R\|^2] = 2B \mathbb{E}_{\mu_s}[R] \leq 2BR_1,$$

using monotonicity together with Lemma 2 for the last step. Hence $|f'(s)| \leq L_f$ on $[\tau_1, \tau_Q]$, so f is L_f -Lipschitz there and the left-Riemann-sum error on panel i is at most $\frac{L_f}{2}(\tau_{i+1} - \tau_i)^2$. Summing,

$$|(A)| \leq \frac{L_f \Delta}{2} \sum_{i < q} (\tau_{i+1} - \tau_i) = \frac{L_f \Delta}{2} (\tau_q - \tau_1) \leq \frac{L_f \Delta}{2} (\tau_Q - \tau_1).$$

Step 3 (term B). Fix i and let

$$\delta_i := \frac{d\mu_{\tau_i}^{K(\tau_i)}}{d\mu_{\tau_i}} - 1, \quad \mathbb{E}_{\mu_{\tau_i}}[\delta_i] = 0, \quad \mathbb{E}_{\mu_{\tau_i}}[\delta_i^2] = \chi^2(\mu_{\tau_i}^{K(\tau_i)} \parallel \mu_{\tau_i}) \leq \varepsilon_\star^2$$

by construction of $K(\cdot)$. Since

$$\mathbb{E}_{\mu_{\tau_i}}[R^2] = \mathbb{V}_{\mu_{\tau_i}}[R] + \mathbb{E}_{\mu_{\tau_i}}[R]^2 \leq 2BR_1 + R_1^2 = \tilde{R}_2,$$

Cauchy–Schwarz gives

$$|\mathbb{E}_{\mu_{\tau_i}^{K(\tau_i)}}[R] - \mathbb{E}_{\mu_{\tau_i}}[R]| = |\mathbb{E}_{\mu_{\tau_i}}[R \delta_i]| \leq \sqrt{\mathbb{E}_{\mu_{\tau_i}}[R^2] \mathbb{E}_{\mu_{\tau_i}}[\delta_i^2]} \leq \sqrt{\tilde{R}_2} \varepsilon_\star.$$

Summing with weights, we find $|(B)| \leq \frac{1}{2} \sqrt{\tilde{R}_2} \varepsilon_\star (\tau_Q - \tau_1)$.

Step 4 (term C). For a given i , write $\bar{R} := \mathbb{E}_{\mu_{\tau_i}}[R]$. Since $\mathbb{E}_{\mu_{\tau_i}}[\delta_i] = 0$, subtracting the constant $\bar{R}$ before applying Cauchy–Schwarz does not change the identity, and it minimizes the resulting bound:

$$\begin{aligned} \mathbb{E}_{\mu_{\tau_i}^{K(\tau_i)}}[(R - \bar{R})^2] &= \mathbb{E}_{\mu_{\tau_i}}[(R - \bar{R})^2(1 + \delta_i)] \\ &= \mathbb{V}_{\mu_{\tau_i}}[R] + \mathbb{E}_{\mu_{\tau_i}}[(R - \bar{R})^2 \delta_i] \\ &\leq \mathbb{V}_{\mu_{\tau_i}}[R] + \sqrt{\mathbb{E}_{\mu_{\tau_i}}[(R - \bar{R})^4] \varepsilon_\star} \end{aligned}$$

by Cauchy–Schwarz. Using the Poincaré bound $\mathbb{V}_{\mu_{\tau_i}}[R] \leq 2BR_1$ and Lemma 5,

$$\mathbb{E}_{\mu_{\tau_i}^{K(\tau_i)}}[(R - \bar{R})^2] \leq 2BR_1 + \sqrt{\widehat{R}_4} \varepsilon_\star.$$

Since variance is the minimizer of $c \mapsto \mathbb{E}_{\mu_{\tau_i}^{K(\tau_i)}}[(R - c)^2]$,

$$\mathbb{V}_{\mu_{\tau_i}^{K(\tau_i)}}[R] \leq \mathbb{E}_{\mu_{\tau_i}^{K(\tau_i)}}[(R - \bar{R})^2] \leq 2BR_1 + \sqrt{\widehat{R}_4} \varepsilon_\star =: \widehat{V}.$$

The draws $L_1^{(i)}, \dots, L_M^{(i)}$ are i.i.d. from $\mu_{\tau_i}^{K(\tau_i)}$ and, crucially, independent across i , since Algorithm 2 draws a fresh batch of M chains at every grid point. Define

$$X_i := \widehat{R}_M(\tau_i) - \mathbb{E}_{\mu_{\tau_i}^{K(\tau_i)}}[R] \quad \Rightarrow \quad \mathbb{E}[X_i] = 0, \mathbb{V}[X_i] \leq \frac{\widehat{V}}{M}.$$

What Step 0 actually needs is $\mathbb{E}[\max_q |(C)_q|]$, not $\max_q \mathbb{E}|(C)_q|$, so rather than summing panel-wise, we define

$$S_q := \sum_{i < q} (\tau_{i+1} - \tau_i) X_i, \quad (C)_q = S_q/2,$$

and note that $\{S_q\}$ is a martingale in q (each X_i is independent of, and mean zero given, $X_1, \dots, X_{i-1}$). Applying Doob's L^2 maximal inequality,

$$\mathbb{E}[\max_q |S_q|] \leq 2\sqrt{\mathbb{V}[S_Q]} = 2\sqrt{\sum_{i < Q} (\tau_{i+1} - \tau_i)^2 \mathbb{V}[X_i]} \leq 2\sqrt{\frac{\widehat{V} \Delta (\tau_Q - \tau_1)}{M}},$$

because $(\tau_{i+1} - \tau_i) \leq \Delta$ and $\sum_{i < Q} (\tau_{i+1} - \tau_i) = \tau_Q - \tau_1$. Hence

$$\mathbb{E}[\max_q |(C)_q|] \leq \sqrt{\frac{\widehat{V} \Delta (\tau_Q - \tau_1)}{M}}.$$

Step 5 (assemble). Terms (A) and (B) are bounded surely for every q at once (Steps 2–3); only (C) needed Step 4's maximal inequality. Combining with Step 0,

$$\mathbb{E}[W_1(\nu_\star^Q, \nu_\star^{Q, \text{TI}})] \leq \frac{\tau_Q - \tau_1}{2} \left(\frac{\tau_Q - \tau_1}{2} (L_f \Delta + \sqrt{\widetilde{R}_2} \varepsilon_\star) + \sqrt{\frac{\widehat{V} \Delta (\tau_Q - \tau_1)}{M}} \right),$$

which is the claimed bound. $\square$

C Proofs of the main results

C.1 Proof of Theorem 1

By the triangle inequality and $(a + b)^2 \leq 2a^2 + 2b^2$,

$$\begin{aligned} & \text{TV}^2 \left(\int \mu_\tau \nu_\star(d\tau), \int \mu_\tau^{K'} \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right) \\ & \leq 2\text{TV}^2 \left(\int \mu_\tau \nu_\star(d\tau), \int \mu_\tau \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right) \\ & \quad + 2\text{TV}^2 \left(\int \mu_\tau \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau), \int \mu_\tau^{K'} \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right). \end{aligned}$$

By the convergence guarantee from [Andrieu et al. \(2024\)](#) and the Pinsker-like inequality for chi-squared divergence, we have

$$\text{TV}^2(\mu_\tau, \mu_\tau^{K(\tau)}) \leq \frac{1}{2} \chi^2(\mu_\tau^{K(\tau)} \parallel \mu_\tau) \leq \frac{1}{2} \varepsilon_\star^2,$$

so by definition of total variation distance

$$\begin{aligned} & \text{TV} \left(\int \mu_\tau \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau), \int \mu_\tau^{K'} \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right) \\ &= \sup_{|f| \leq 1} \frac{1}{2} \left| \int \left(\int f d\mu_\tau - \int f d\mu_\tau^{K'} \right) \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right| \\ &\leq \sup_{|f| \leq 1} \int \frac{1}{2} \left| \int f d\mu_\tau - \int f d\mu_\tau^{K'} \right| \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \\ &\leq \int \text{TV}(\mu_\tau, \mu_\tau^{K'}) \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \\ &\leq \frac{1}{\sqrt{2}} \varepsilon_\star. \end{aligned}$$

The other term requires some Lipschitz analysis. By Pinsker's inequality, for any τ and τ' ,

$$\begin{aligned} \text{TV}^2(\mu_\tau, \nu_{\tau'}) &\leq \frac{1}{2} \text{KL}(\nu_{\tau'} \parallel \mu_\tau) \\ &= \frac{1}{4} (\tau - \tau') \mathbb{E}_{\nu_{\tau'}}[R(\mathbf{L})] + \frac{1}{2} (\log Z_{\mu_\tau} - \log Z_{\nu_{\tau'}}) \\ &= \frac{1}{4} (\tau - \tau') \mathbb{E}_{\nu_{\tau'}}[R(\mathbf{L})] + \frac{1}{2} \int_\tau^{\tau'} \frac{1}{2} \mathbb{E}_{\nu_t}[R(\mathbf{L})] dt \\ &\leq \frac{1}{2} |\tau - \tau'| \mathbb{E}_\mu[R(\mathbf{L})]. \end{aligned}$$

The last step is because $\mathbb{E}_{\nu_t}[R(\mathbf{L})]$ is decreasing in t , so we can set $t = 0$ for an upper bound. By convexity of TV,

$$\text{TV} \left(\int \mu_\tau \nu_\star(d\tau), \int \mu_\tau \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right) \leq \int \text{TV}(\mu_\tau, \nu_{\tau'}) \gamma(d\tau, d\tau')$$

for $\gamma \in \Gamma(\nu_\star, \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}])$, so

$$\begin{aligned} \text{TV} \left(\int \mu_\tau \nu_\star(d\tau), \int \mu_\tau \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right) &\leq \int \sqrt{\frac{1}{2} |\tau - \tau'| \mathbb{E}_\mu[R(\mathbf{L})]} \gamma(d\tau, d\tau') \quad \forall \gamma \\ &\leq \sqrt{\int \frac{1}{2} |\tau - \tau'| \mathbb{E}_\mu[R(\mathbf{L})]} \gamma(d\tau, d\tau') \quad \forall \gamma \\ \Rightarrow \text{TV} \left(\int \mu_\tau \nu_\star(d\tau), \int \mu_\tau \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right) &\leq \sqrt{\frac{1}{2} \mathbb{E}_\mu[R(\mathbf{L})]} W_1(\nu_\star, \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}]). \end{aligned}$$

We can bound the residual term by Lemma 2, so now we bound the W_1 term. We can see that

$$W_1(\nu_\star, \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}]) \leq \mathbb{E}[W_1(\nu_\star, \nu_\star^{Q, \text{TI}})]$$

by Jensen's inequality, and

$$\mathbb{E}[W_1(\nu_\star, \nu_\star^{Q, \text{TI}})] \leq W_1(\nu_\star, \nu_\star^Q) + \mathbb{E}[W_1(\nu_\star^Q, \nu_\star^{Q, \text{TI}})]$$

by the triangle inequality. The desired result follows from Lemmas 4 and 6.

C.2 Proof of Theorem 2

By the triangle inequality,

$$\begin{aligned} & W_1 \left(\int \mu_\tau \nu_\star(d\tau), \int \mu_\tau^{K'} \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right) \\ & \leq W_1 \left(\int \mu_\tau \nu_\star(d\tau), \int \mu_\tau \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right) \\ & \quad + W_1 \left(\int \mu_\tau \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau), \int \mu_\tau^{K'} \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right). \end{aligned}$$

By the convergence guarantee from [Andrieu et al. \(2024\)](#) and the Talagrand's T1 inequality, we have

$$W_1^2(\mu_\tau, \mu_\tau^{K(\tau)}) \leq B\chi^2(\mu_\tau^{K(\tau)} \parallel \mu_\tau) \leq B\varepsilon_\star^2,$$

so by definition of Wasserstein distance

$$\begin{aligned} & W_1 \left(\int \mu_\tau \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau), \int \mu_\tau^{K'} \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right) \\ & = \sup_{\text{Lip}(f) < 1} \left| \int \left(\int f d\mu_\tau - \int f d\mu_\tau^{K'} \right) \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right| \\ & \leq \sup_{\text{Lip}(f) < 1} \int \left| \int f d\mu_\tau - \int f d\mu_\tau^{K'} \right| \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \\ & \leq \int W_1(\mu_\tau, \mu_\tau^{K'}) \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \\ & \leq \sqrt{B} \varepsilon_\star. \end{aligned}$$

The other term requires some Lipschitz analysis. By Talagrand's T1 inequality, for any τ and τ' ,

$$\begin{aligned} W_1^2(\mu_\tau, \nu_{\tau'}) & \leq \frac{1}{2} \text{KL}(\nu_{\tau'} \parallel \mu_\tau) \\ & = \frac{1}{4}(\tau - \tau') \mathbb{E}_{\nu_{\tau'}}[R(\mathbf{L})] + \frac{1}{2} (\log Z_{\mu_\tau} - \log Z_{\nu_{\tau'}}) \\ & = \frac{1}{4}(\tau - \tau') \mathbb{E}_{\nu_{\tau'}}[R(\mathbf{L})] + \frac{1}{2} \int_\tau^{\tau'} \frac{1}{2} \mathbb{E}_{\nu_t}[R(\mathbf{L})] dt \\ & \leq \frac{1}{2} |\tau - \tau'| \mathbb{E}_\mu[R(\mathbf{L})]. \end{aligned}$$

The last step is because $\mathbb{E}_{\nu_t}[R(\mathbf{L})]$ is decreasing in t , so we can set $t = 0$ for an upper bound. By convexity of W_1^2 ,

$$W_1^2 \left(\int \mu_\tau \nu_\star(d\tau), \int \mu_\tau \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right) \leq \int W_1^2(\mu_\tau, \nu_{\tau'}) \gamma(d\tau, d\tau')$$

for $\gamma \in \Gamma(\nu_\star, \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}])$, so

$$\begin{aligned} W_1 \left(\int \mu_\tau \nu_\star(d\tau), \int \mu_\tau \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right) & \leq \int \sqrt{B|\tau - \tau'| \mathbb{E}_\mu[R(\mathbf{L})] \gamma(d\tau, d\tau')} \quad \forall \gamma \\ & \leq \sqrt{\int B|\tau - \tau'| \mathbb{E}_\mu[R(\mathbf{L})] \gamma(d\tau, d\tau')} \quad \forall \gamma \\ \Rightarrow W_1 \left(\int \mu_\tau \nu_\star(d\tau), \int \mu_\tau \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}](d\tau) \right) & \leq \sqrt{B \mathbb{E}_\mu[R(\mathbf{L})] W_1(\nu_\star, \mathbb{E}_{\mu^K}[\nu_\star^{Q, \text{TI}}])}. \end{aligned}$$

We can bound the residual term by Lemma 2, so now we bound the W_1 term. We can see that

$$W_1(\nu_*, \mathbb{E}_{\mu^\kappa}[\nu_*^{Q, \text{TI}}]) \leq \mathbb{E}[W_1(\nu_*, \nu_*^{Q, \text{TI}})]$$

by Jensen's inequality, and

$$\mathbb{E}[W_1(\nu_*, \nu_*^{Q, \text{TI}})] \leq W_1(\nu_*, \nu_*^Q) + \mathbb{E}[W_1(\nu_*^Q, \nu_*^{Q, \text{TI}})]$$

by the triangle inequality. The desired result follows from Lemmas 4 and 6.

C.3 Proof of Corollary 1

$\lambda = \Theta(n_1^a)$ and $B = \Theta(1)$ imply

$$\log(u_0(t) + 1) = \Theta\left(n_1^{1/2-a} \sqrt{A(t)} + n_1^{1-2a}\right),$$

and because $A(t) = \Theta(n_1 n_2)$ under our standardization regime, we have

$$\log \log(u_0(t)/2) = \log(\Theta(n_1^{1-a} \sqrt{n_2})) = \Theta(\log(n_1 n_2)).$$

It follows that $K(t) = \tilde{O}(n_1^{2-2a} n_2)$. To get our overall TV distance down to $\eta < 1$, because of the fact that the M -term depends only on the product QM while the other terms depend only on Q , we set $M = 1$. Furthermore, the M -term is the slowest to decline, so to find the Q needed to bring us to η , we choose

$$\sqrt{\frac{N}{Q}} = \Omega\left(\frac{\eta^2}{N}\right) \Rightarrow Q = O\left(\frac{N^3}{\eta^4}\right).$$

If $\varepsilon_* = \Theta(\eta)$, it follows that the number of RWM steps needed is

$$QMK = \tilde{O}\left(\frac{N^3 n_1^{2-2a} n_2}{\eta^4}\right).$$

C.4 Proof of Corollary 2

If $B = \Theta(1)$, the convergence result of Theorem 2 is the same in big-O terms as that of Theorem 1, so we can simply replace η with $n_1 n_2 \eta$ in the result of Corollary 1 and obtain

$$QMK = \tilde{O}\left(\frac{N^3 n_1^{2-2a} n_2}{n_1^4 n_2^4 \eta^4}\right).$$

Because $N \leq n_1 n_2$, we can also say

$$QMK = \tilde{O}\left(\frac{n_1^{1-2a}}{\eta^4}\right).$$